

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv(r"C:\Users\304690\Downloads\List of Countries by Sugarcane Production.csv")

df.head()

```

	Unnamed: 0	Country	Continent	Production (Tons) \
0	0	Brazil	South America	768.678.382
1	1	India	Asia	348.448.000
2	2	China	Asia	123.059.739
3	3	Thailand	Asia	87.468.496
4	4	Pakistan	Asia	65.450.704

	Production per Person (Kg)	Acreage (Hectare)	Yield (Kg / Hectare)
0	3.668,531	10.226.205	75.167,5
1	260721	4.950.000	70.393,5
2	88287	1.675.215	73.459,1
3	1.264,303	1.336.575	65.442,2
4	324219	1.130.820	57.879

```

df.shape

(103, 7)

```

Data Cleaning

```

#Replacing . with whitespace and , with .
df['Production (Tons)']=df['Production (Tons)'].str.replace('.', '')
df['Production per Person (Kg)']=df['Production per Person (Kg)'].str.replace('.', '').str.replace(',','.')
df['Acreage (Hectare)']=df['Acreage (Hectare)'].str.replace('.', '')
df['Yield (Kg / Hectare)']=df['Yield (Kg / Hectare)'].str.replace('.', '').str.replace(',','.')

df.head()

```

	Unnamed: 0	Country	Continent	Production (Tons) \
0	0	Brazil	South America	768678382
1	1	India	Asia	348448000
2	2	China	Asia	123059739
3	3	Thailand	Asia	87468496
4	4	Pakistan	Asia	65450704

	Production per Person (Kg)	Acreage (Hectare)	Yield (Kg / Hectare)
0	3668.531	10226205	75167.5

1	260721	4950000	70393.5
2	88287	1675215	73459.1
3	1264.303	1336575	65442.2
4	324219	1130820	57879

#Renaming column

```
df.rename(columns={'Production (Tons)': 'Production(Tons)', 'Production
per Person (Kg)': 'Production_per_Person', 'Acreage
(Hectare)': 'Acreage(Hectare)', 'Yield (Kg /
Hectare)': 'Yield(kg/Hectare)'}, inplace=True)
```

df

	Unnamed: 0	Country	Continent	Production(Tons)	\
0	0	Brazil	South America	768678382	
1	1	India	Asia	348448000	
2	2	China	Asia	123059739	
3	3	Thailand	Asia	87468496	
4	4	Pakistan	Asia	65450704	
...	...	...	...	...	
98	98	Lebanon	Asia	97	
99	99	Djibouti	Africa	53	
100	100	Singapore	Asia	50	
101	101	Samoa	Oceania	12	
102	102	Syria	Asia	1	

	Production_per_Person	Acreage(Hectare)	Yield(kg/Hectare)
0	3668.531	10226205	75167.5
1	260721	4950000	70393.5
2	88287	1675215	73459.1
3	1264.303	1336575	65442.2
4	324219	1130820	57879
...	...	...	...
98	16	3	28386.4
99	51	NaN	NaN
100	9	2	25
101	6	1	11949.8
102	0	0	83034.2

[103 rows x 7 columns]

#Checking null value

```
df.isnull().sum()
```

Unnamed: 0	0
Country	0
Continent	0
Production(Tons)	0
Production_per_Person	0
Acreage(Hectare)	1

```
Yield(kg/Hectare      1
dtype: int64
```

```
df[df['Acreage(Hectare)'].isnull()]
```

```
   Unnamed: 0  Country Continent Production(Tons)
Production_per_Person \
99           99  Djibouti      Africa           53
51
```

```
   Acreage(Hectare) Yield(kg/Hectare
99              NaN              NaN
```

```
df=df.dropna().reset_index()
df
```

```
   index  Unnamed: 0  Country Continent Production(Tons) \
0        0          0   Brazil  South America      768678382
1        1          1    India         Asia      348448000
2        2          2    China         Asia      123059739
3        3          3  Thailand         Asia      87468496
4        4          4  Pakistan         Asia      65450704
..      ...        ...      ...      ...      ...
97       97         97    Spain      Europe        394
98       98         98   Lebanon         Asia         97
99      100        100  Singapore         Asia         50
100     101        101    Samoa      Oceania         12
101     102        102    Syria         Asia          1
```

```
   Production_per_Person  Acreage(Hectare)  Yield(kg/Hectare
0                3668.531         10226205         75167.5
1                260721         4950000         70393.5
2                88287         1675215         73459.1
3               1264.303         1336575         65442.2
4               324219         1130820         57879
..      ...        ...      ...      ...
97                8          9         43596.5
98               16          3         28386.4
99                9          2           25
100               6          1         11949.8
101               0          0         83034.2
```

```
[102 rows x 8 columns]
```

```
df.drop(['Unnamed: 0', 'index'], inplace=True, axis=1)
```

```
df
```

```
   Country      Continent Production(Tons)
Production_per_Person \
0      Brazil  South America      768678382      3668.531
```

1	India	Asia	348448000	260721
2	China	Asia	123059739	88287
3	Thailand	Asia	87468496	1264.303
4	Pakistan	Asia	65450704	324219
..	...	...	...	...
97	Spain	Europe	394	8
98	Lebanon	Asia	97	16
99	Singapore	Asia	50	9
100	Samoa	Oceania	12	6
101	Syria	Asia	1	0

	Acreage(Hectare)	Yield(kg/Hectare)
0	10226205	75167.5
1	4950000	70393.5
2	1675215	73459.1
3	1336575	65442.2
4	1130820	57879
..	...	...
97	9	43596.5
98	3	28386.4
99	2	25
100	1	11949.8
101	0	83034.2

[102 rows x 6 columns]

df.dtypes

Country	object
Continent	object
Production(Tons)	object
Production_per_Person	object
Acreage(Hectare)	object
Yield(kg/Hectare)	object
dtype:	object

Datatype Conversion

```
df['Yield(kg/Hectare)']=df['Yield(kg/Hectare)'].astype('float')
df['Production_per_Person']=df['Production_per_Person'].astype('float')
)
```

```
df['Production(Tons)']=df['Production(Tons)'].astype('float')
df['Acreage(Hectare)']=df['Acreage(Hectare)'].astype('float')
df.dtypes
```

```
Country          object
Continent        object
Production(Tons)  float64
Production_per_Person  float64
Acreage(Hectare)  float64
Yield(kg/Hectare  float64
dtype: object
```

```
df.nunique()
```

```
Country          102
Continent         6
Production(Tons)  102
Production_per_Person  101
Acreage(Hectare)  101
Yield(kg/Hectare  102
dtype: int64
```

```
# Checking Duplicacy
df.duplicated().sum()
```

```
0
```

Univariate Analysis

```
df.head()
```

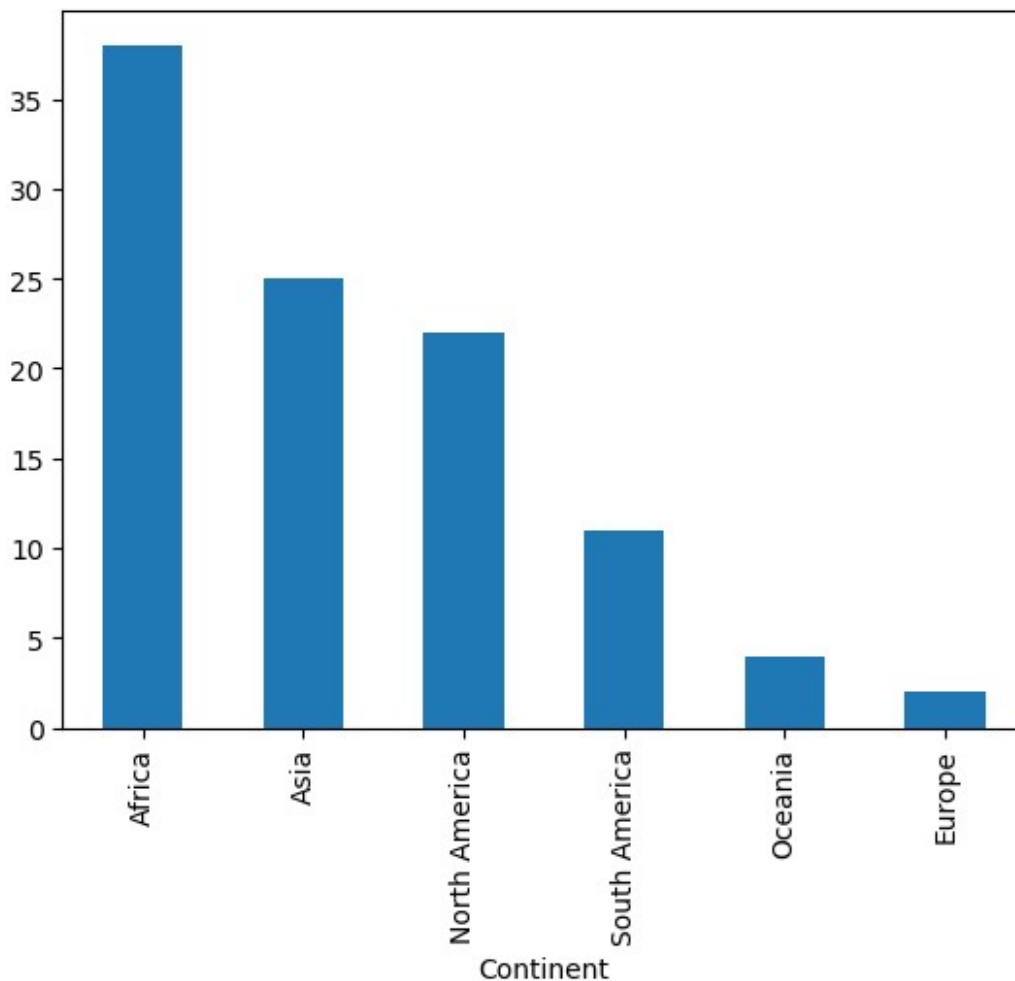
	Country	Continent	Production(Tons)	Production_per_Person \
0	Brazil	South America	768678382.0	3668.531
1	India	Asia	348448000.0	260721.000
2	China	Asia	123059739.0	88287.000
3	Thailand	Asia	87468496.0	1264.303
4	Pakistan	Asia	65450704.0	324219.000

	Acreage(Hectare)	Yield(kg/Hectare)
0	10226205.0	75167.5
1	4950000.0	70393.5
2	1675215.0	73459.1
3	1336575.0	65442.2
4	1130820.0	57879.0

Q1) How many countries produce sugarcane from each continent

```
df['Continent'].value_counts().plot(kind='bar')
```

```
<Axes: xlabel='Continent'>
```



```
# Representing through distribution
plt.figure(figsize=(10,10))
plt.subplot(2,2,1)
sns.distplot(df['Production(Tons)'])
plt.subplot(2,2,2)
sns.distplot(df['Production_per_Person'])
plt.subplot(2,2,3)
sns.distplot(df['Acreage(Hectare)'])
plt.subplot(2,2,4)
sns.distplot(df['Yield(kg/Hectare)'])
plt.show()
```

```
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2848997698.py:3:  
UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn  
v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['Production(Tons)'])  
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2848997698.py:5:  
UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn  
v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['Production_per_Person'])  
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2848997698.py:7:  
UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn  
v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

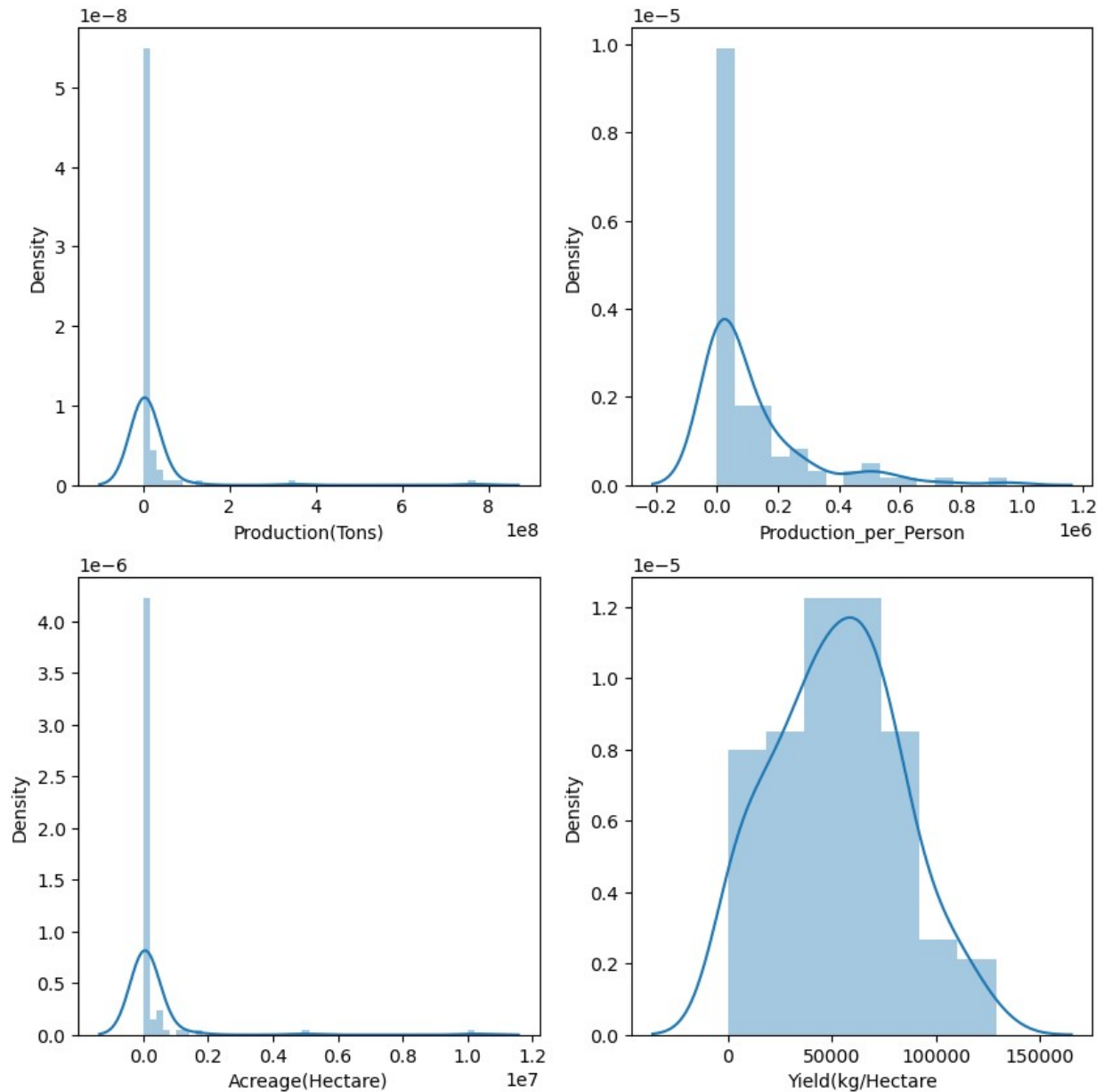
```
sns.distplot(df['Acreage(Hectare)'])  
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2848997698.py:9:  
UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn  
v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

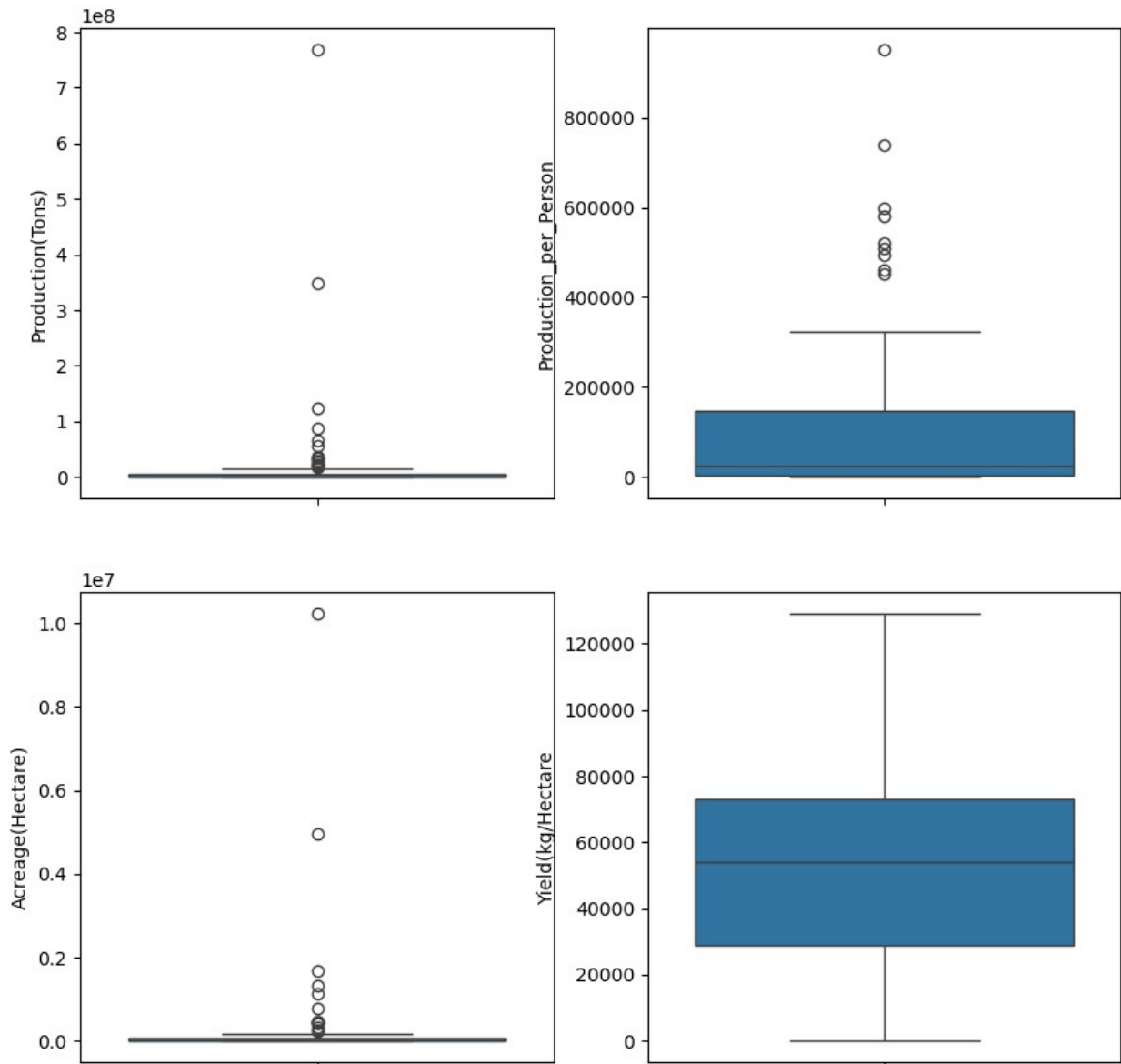
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['Yield(kg/Hectare)'])
```



```
# Understanding the outlier through Boxplot  
plt.figure(figsize=(10,10))
```

```
plt.subplot(2,2,1)
sns.boxplot(df['Production(Tons)'])
plt.subplot(2,2,2)
sns.boxplot(df['Production_per_Person'])
plt.subplot(2,2,3)
sns.boxplot(df['Acreage(Hectare)'])
plt.subplot(2,2,4)
sns.boxplot(df['Yield(kg/Hectare)'])
plt.show()
```



```
df.describe()
```

	Production(Tons)	Production_per_Person	Acreage(Hectare)	\
count	1.020000e+02	102.000000	1.020000e+02	
mean	1.850372e+07	112952.435755	2.498981e+05	
std	8.419149e+07	176651.341929	1.137003e+06	
min	1.000000e+00	0.000000	0.000000e+00	
25%	6.251875e+04	3671.910000	1.104000e+03	
50%	1.440044e+06	25572.500000	1.655800e+04	
75%	6.426824e+06	146384.750000	8.047400e+04	
max	7.686784e+08	951087.000000	1.022620e+07	

	Yield(kg/Hectare)
count	102.000000
mean	52628.078431
std	30504.676683
min	10.000000
25%	29072.025000
50%	54108.950000
75%	73282.700000
max	129049.300000

Bivariate analysis

```
df.head()
```

	Country	Continent	Production(Tons)	Production_per_Person	\
0	Brazil	South America	768678382.0	3668.531	
1	India	Asia	348448000.0	260721.000	
2	China	Asia	123059739.0	88287.000	
3	Thailand	Asia	87468496.0	1264.303	
4	Pakistan	Asia	65450704.0	324219.000	

	Acreage(Hectare)	Yield(kg/Hectare)
0	10226205.0	75167.5
1	4950000.0	70393.5
2	1675215.0	73459.1
3	1336575.0	65442.2
4	1130820.0	57879.0

Q1) Which country produces maximum sugarcane

```
df_new=df[['Country','Production(Tons)']].set_index('Country')
df_new
```

Country	Production(Tons)
Brazil	768678382.0
India	348448000.0
China	123059739.0
Thailand	87468496.0
Pakistan	65450704.0
...	...
Spain	394.0
Lebanon	97.0
Singapore	50.0
Samoa	12.0
Syria	1.0

[102 rows x 1 columns]

```
df_new['Production(Tons)_percent']=(df_new['Production(Tons)']*100)/
df_new['Production(Tons)'].sum()
```

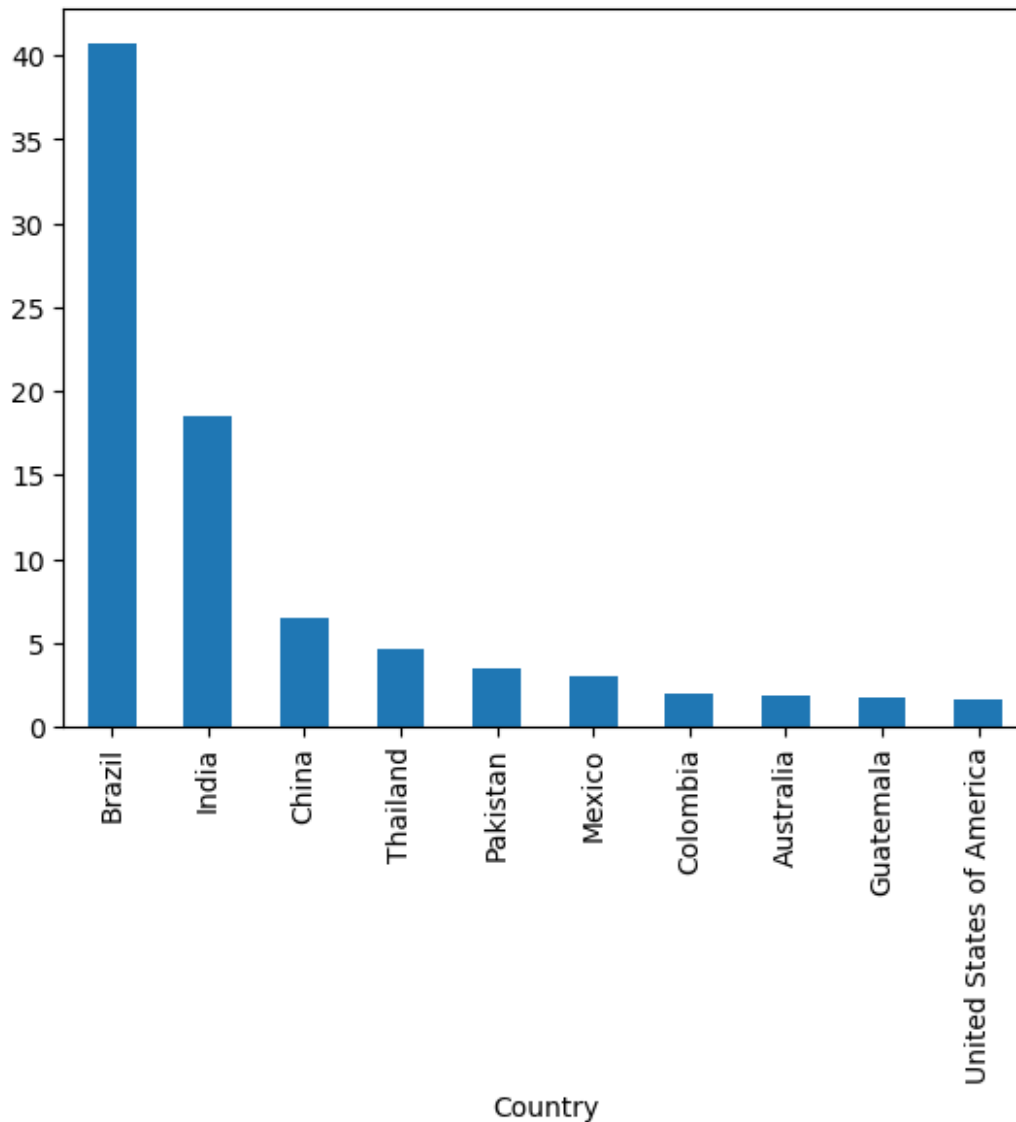
df_new

Country	Production(Tons)	Production(Tons)_percent
Brazil	768678382.0	4.072729e+01
India	348448000.0	1.846200e+01
China	123059739.0	6.520138e+00
Thailand	87468496.0	4.634389e+00
Pakistan	65450704.0	3.467809e+00
...	...	...
Spain	394.0	2.087551e-05
Lebanon	97.0	5.139401e-06
Singapore	50.0	2.649176e-06
Samoa	12.0	6.358022e-07
Syria	1.0	5.298352e-08

[102 rows x 2 columns]

```
df_new['Production(Tons)_percent'].head(10).plot(kind='bar')
```

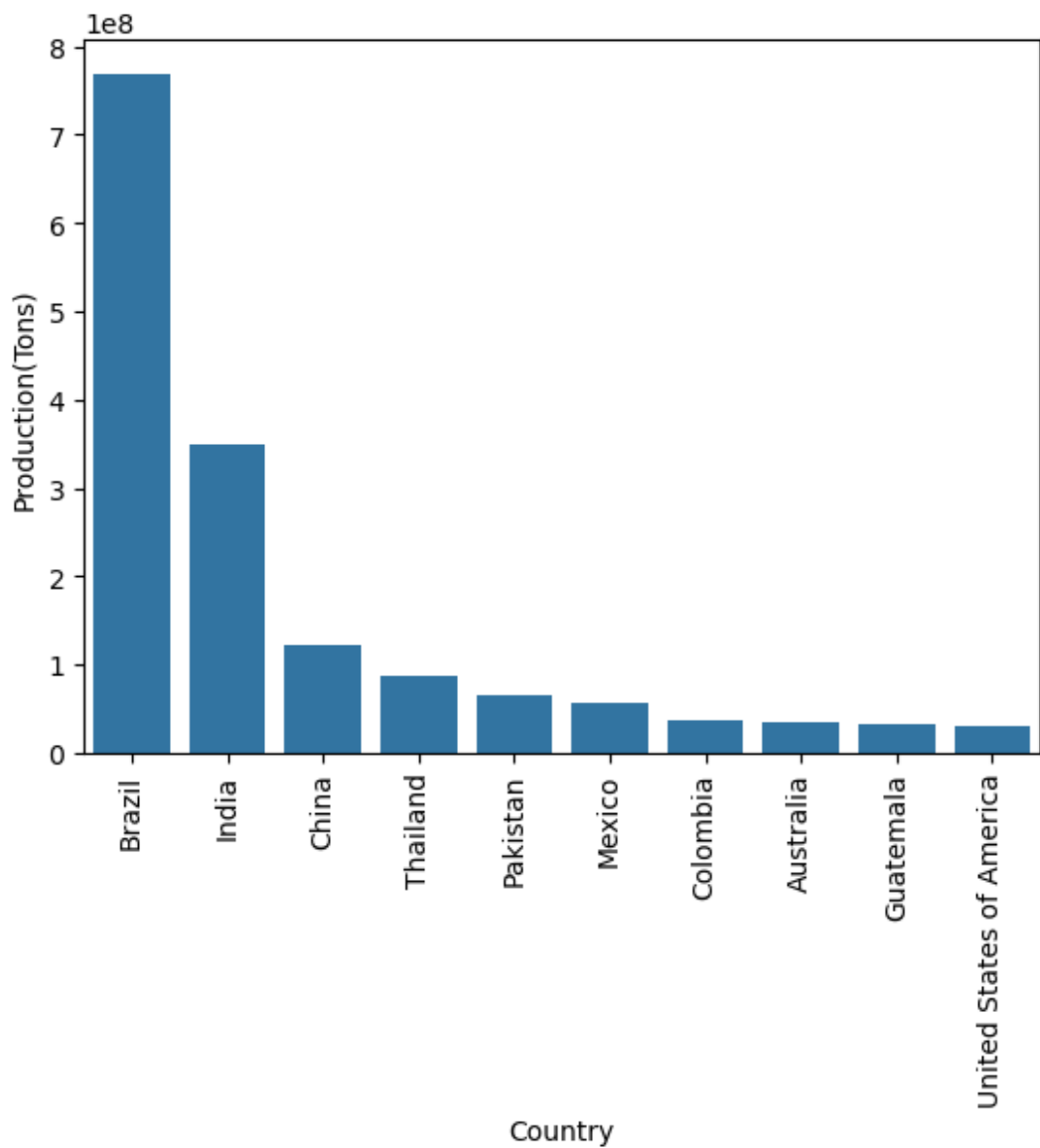
<Axes: xlabel='Country'>



```
# Barplot
```

```
ax=sns.barplot(data=df_new.head(10),x='Country',y='Production(Tons)')  
ax.set_xticklabels(ax.get_xticklabels(),rotation=90)  
plt.show()
```

```
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2033838049.py:3:  
UserWarning: set_ticklabels() should only be used with a fixed number  
of ticks, i.e. after set_ticks() or using a FixedLocator.  
ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
```



Which country has highest land

df.head(10)

	Country	Continent	Production(Tons)	\
0	Brazil	South America	768678382.0	
1	India	Asia	348448000.0	
2	China	Asia	123059739.0	
3	Thailand	Asia	87468496.0	
4	Pakistan	Asia	65450704.0	
5	Mexico	North America	56446821.0	
6	Colombia	South America	36951213.0	
7	Australia	Oceania	34403004.0	

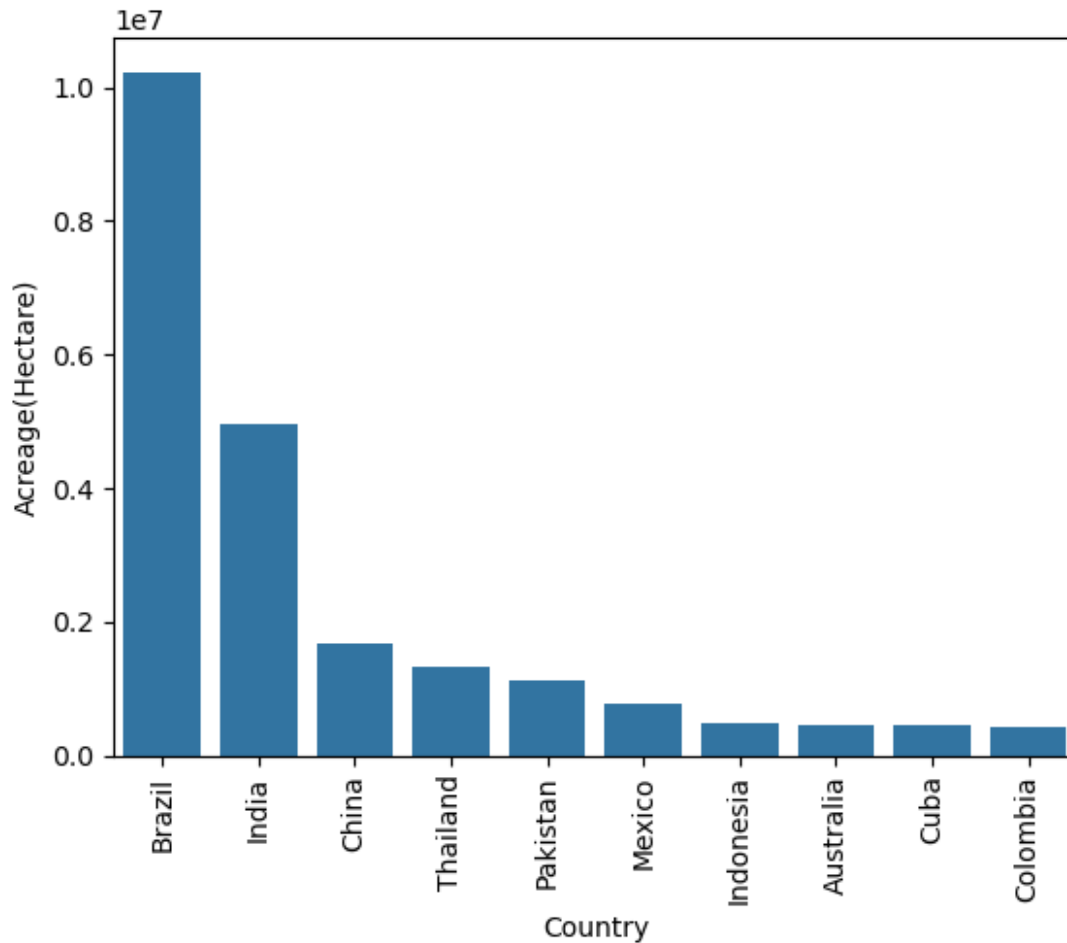
8	Guatemala	North America	33533403.0
9	United States of America	North America	29926210.0

	Production_per_Person	Acreage(Hectare)	Yield(kg/Hectare)
0	3668.531	10226205.0	75167.5
1	260721.000	4950000.0	70393.5
2	88287.000	1675215.0	73459.1
3	1264.303	1336575.0	65442.2
4	324219.000	1130820.0	57879.0
5	452524.000	781054.0	7227.0
6	740075.000	416626.0	88691.5
7	1373.406	447204.0	76929.1
8	1938.114	25985.0	129049.3
9	91304.000	37053.0	80766.0

Barplot

```
df_acr=df.sort_values(by='Acreage(Hectare)',ascending=False)
ax=sns.barplot(data=df_acr.head(10),x='Country',y='Acreage(Hectare)')
ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
plt.show()
```

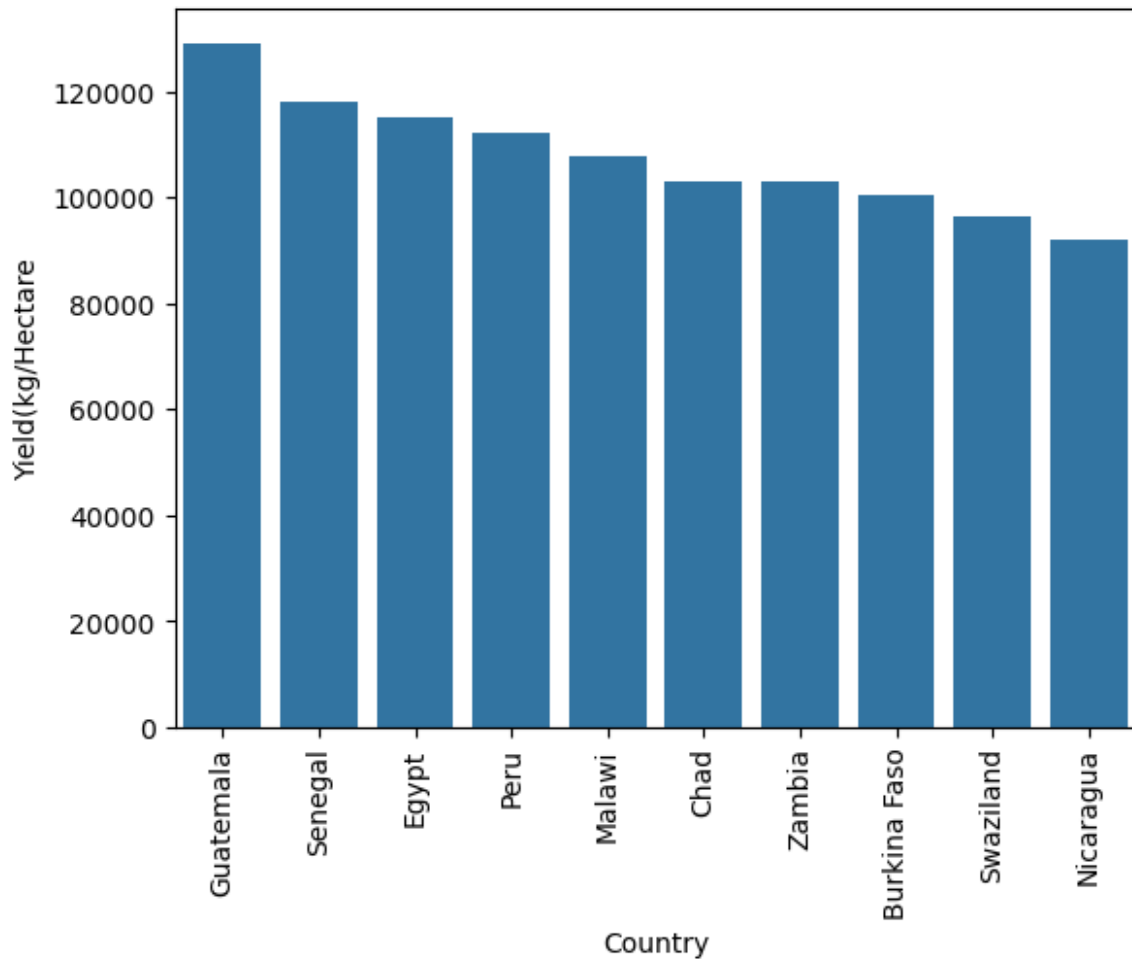
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2159596498.py:4:
 UserWarning: set_ticklabels() should only be used with a fixed number
 of ticks, i.e. after set_ticks() or using a FixedLocator.
 ax.set_xticklabels(ax.get_xticklabels(),rotation=90)



which country has highest yield per hectare

```
df_yield=df.sort_values(by='Yield(kg/Hectare)',ascending=False)
ax=sns.barplot(data=df_yield.head(10),x='Country',y='Yield(kg/Hectare')
)
ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
plt.show()
```

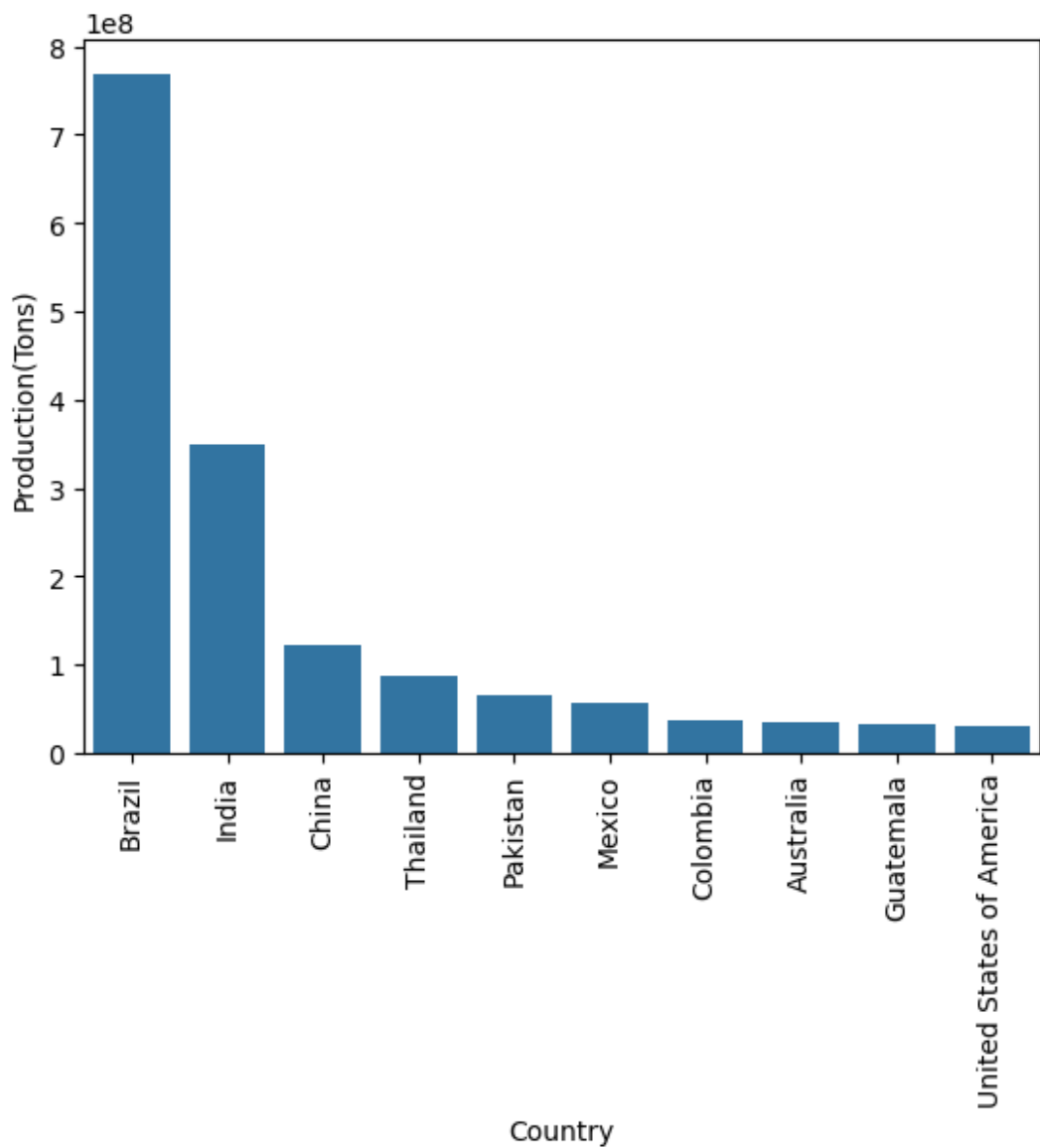
```
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\4214520326.py:3:
UserWarning: set_ticklabels() should only be used with a fixed number
of ticks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
```



Which country has highest production

```
df_pro=df.sort_values(by='Production(Tons)',ascending=False)
ax=sns.barplot(data=df_pro.head(10),x='Country',y='Production(Tons)')
ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
plt.show()
```

```
C:\Users\304690\AppData\Local\Temp\ipykernel_27156\2801629704.py:3:
UserWarning: set_ticklabels() should only be used with a fixed number
of ticks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_xticklabels(ax.get_xticklabels(),rotation=90)
```



Correaltion

```
df.corr(numeric_only=True)
```

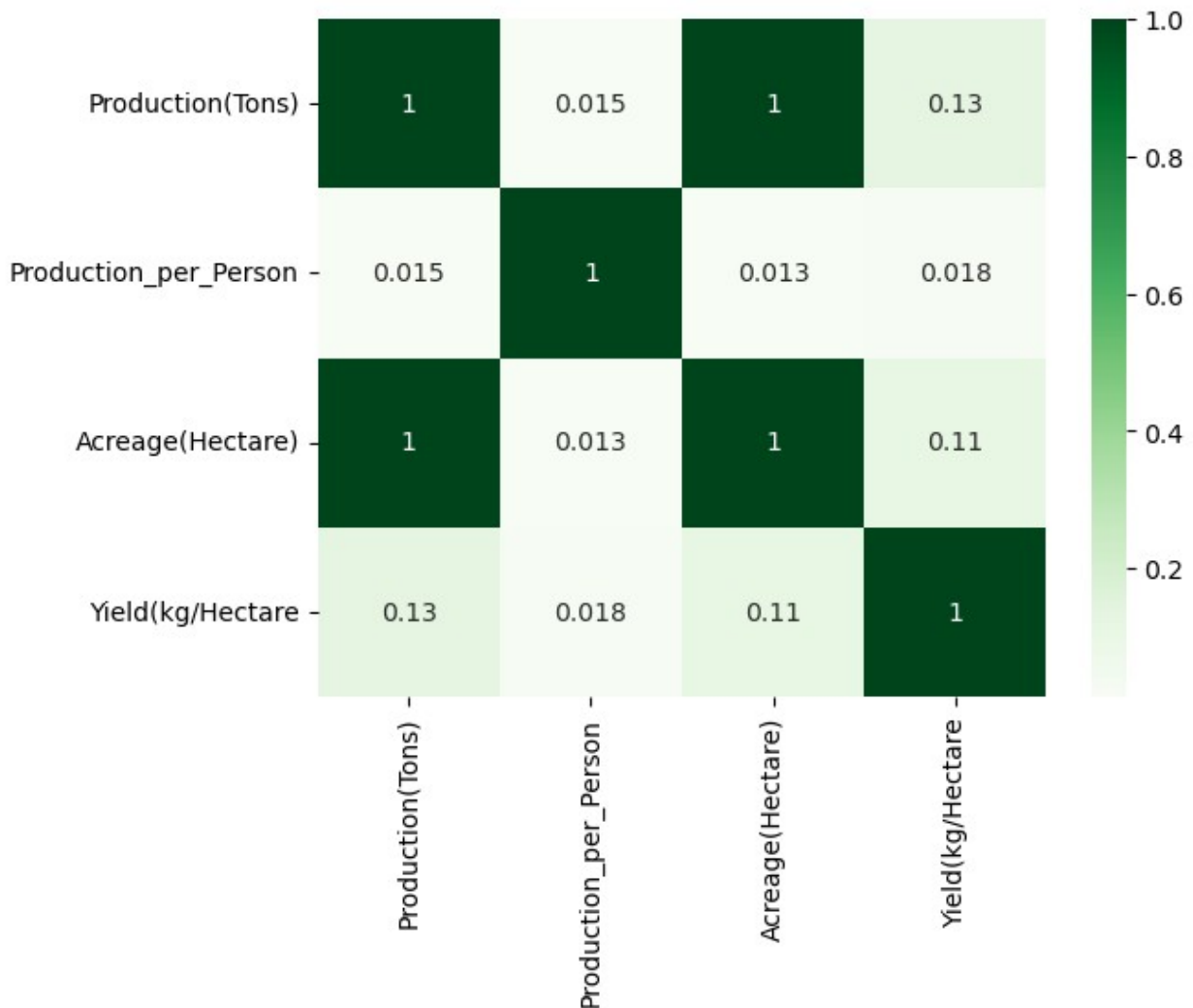
	Production(Tons)	Production_per_Person \
Production(Tons)	1.000000	0.015000
Production_per_Person	0.015000	1.000000
Acreage(Hectare)	0.997550	0.012557
Yield(kg/Hectare)	0.132812	0.017999
	Acreage(Hectare)	Yield(kg/Hectare)
Production(Tons)	0.997550	0.132812
Production_per_Person	0.012557	0.017999

```

Acreage(Hectare)          1.000000      0.113433
Yield(kg/Hectare)         0.113433      1.000000

sns.heatmap(df.corr(numeric_only=True),annot=True,cmap='Greens')
<Axes: >

```

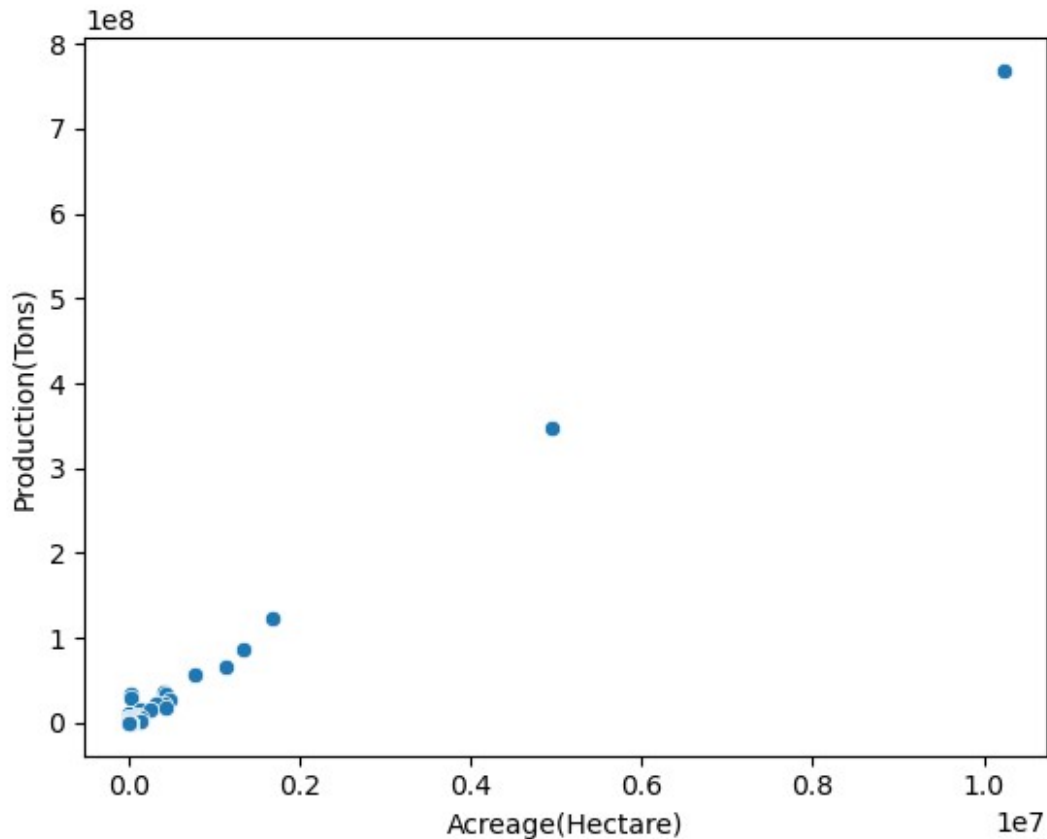


Do country has highest land will produce more sugarcane

```

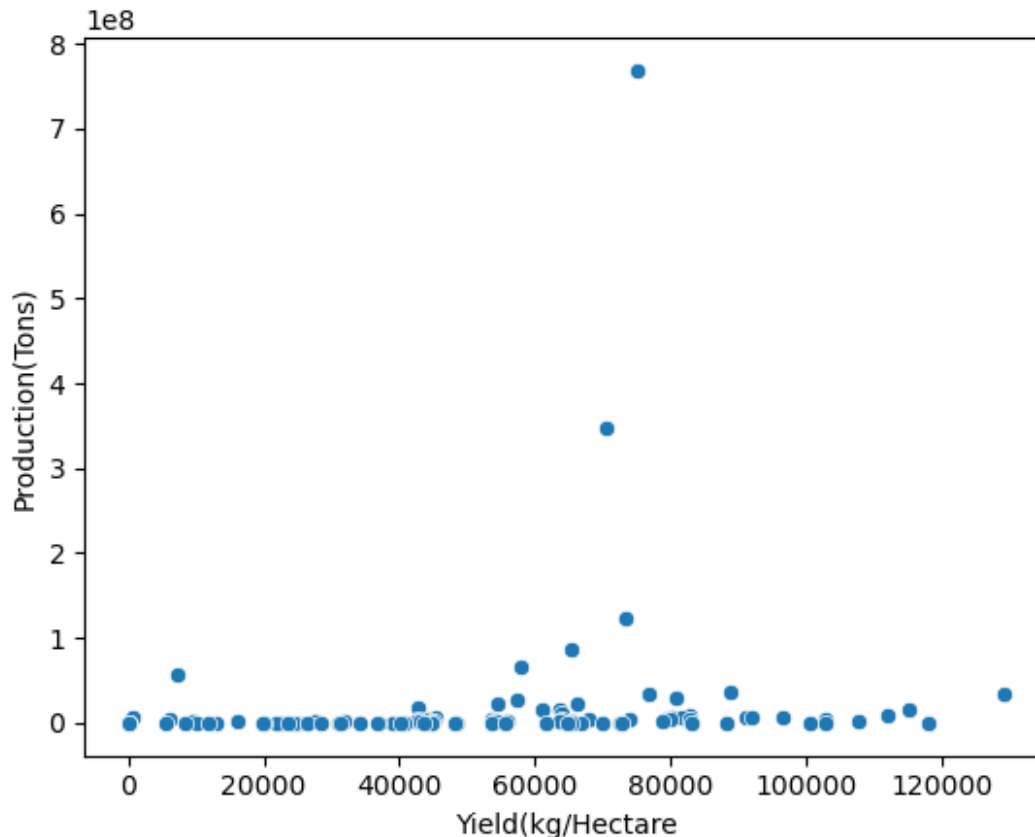
sns.scatterplot(data=df,x='Acreage(Hectare)',y='Production(Tons)')
<Axes: xlabel='Acreage(Hectare)', ylabel='Production(Tons)'>

```



Do country which yield more sugarcane per hectare produce more sugarcane in total

```
sns.scatterplot(data=df,x='Yield(kg/Hectare)',y='Production(Tons)')  
<Axes: xlabel='Yield(kg/Hectare)', ylabel='Production(Tons)'>
```



Analysis the continent

```
df_continent = df.groupby("Continent").sum()
df_continent["number_of_countries"] = df.groupby("Continent").count()[
    "Country"]
df_continent
```

		Country \
Continent		
Africa	EgyptSouth AfricaKenyaSwazilandSudanZambiaMaur...	
Asia	IndiaChinaThailandPakistanIndonesiaPhilippines...	
Europe		PortugalSpain
North America	MexicoGuatemalaUnited States of AmericaCubaEl ...	
Oceania		AustraliaFijiPapua New GuineaSamoa
South America	BrazilColombiaArgentinaPeruEcuadorBoliviaParag...	
Production(Tons)		Production_per_Person
Acreage(Hectare) \		
Continent		
Africa	89681472.0	2332636.293

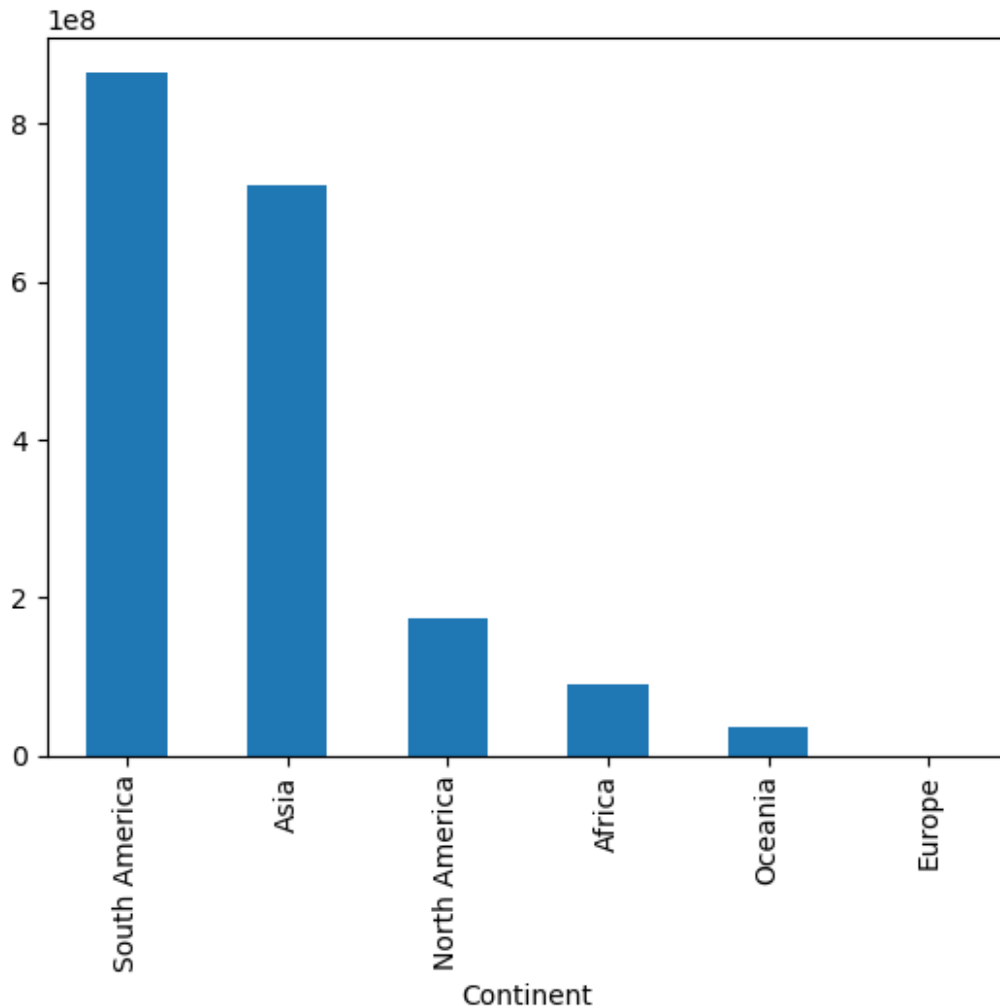
1439089.0		
Asia	721930425.0	1857769.303
10608319.0		
Europe	5823.0	536.000
71.0		
North America	173995947.0	3796081.508
1581983.0		
Oceania	36177574.0	28593.605
490909.0		
South America	865588126.0	3505531.738
11369236.0		

	Yield(kg/Hectare	number_of_countries
Continent		
Africa	2142107.5	38
Asia	1171871.4	25
Europe	131870.9	2
North America	1082602.4	22
Oceania	162419.1	4
South America	677192.7	11

Which continent produce maximum sugarcane

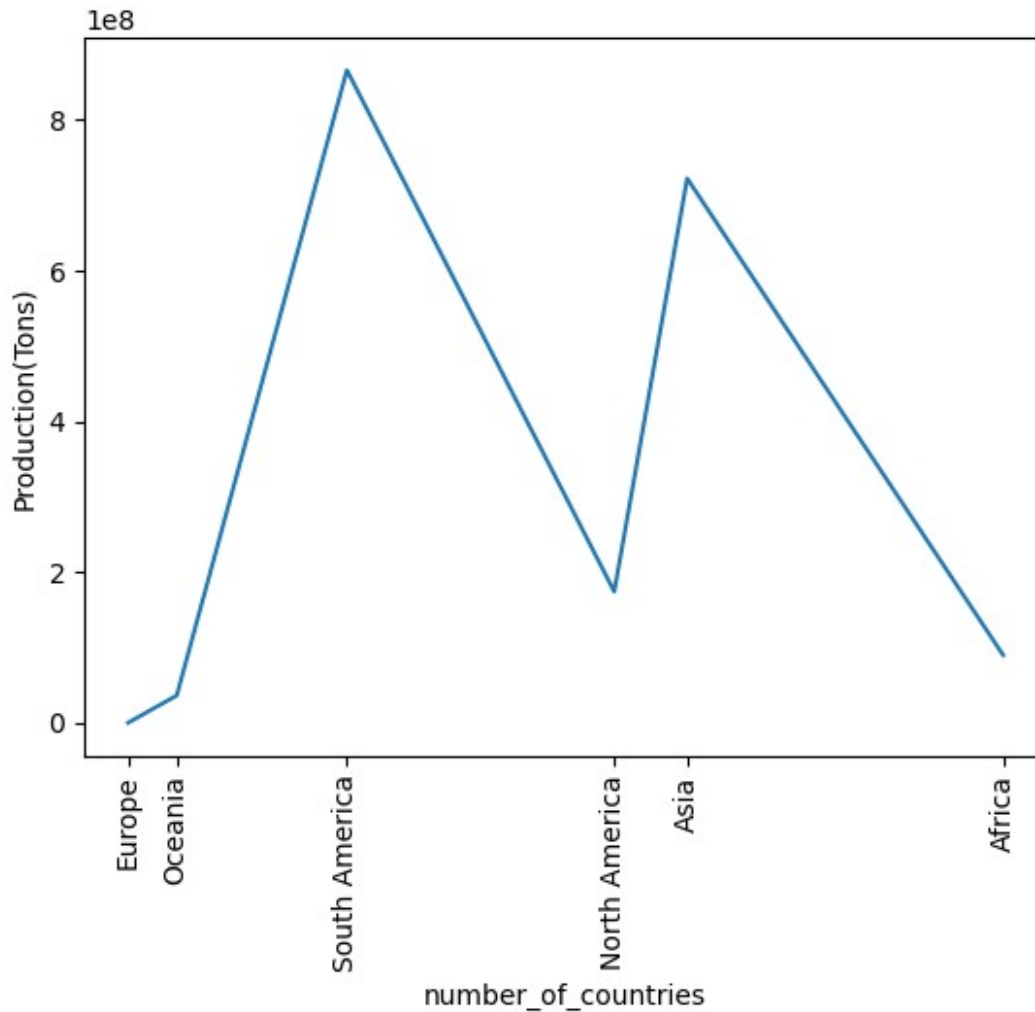
```
df_continent["Production(Tons)"].sort_values(ascending =
False).plot(kind = "bar")

<Axes: xlabel='Continent'>
```



Do number of countries in continent effect the production of sugarcane?

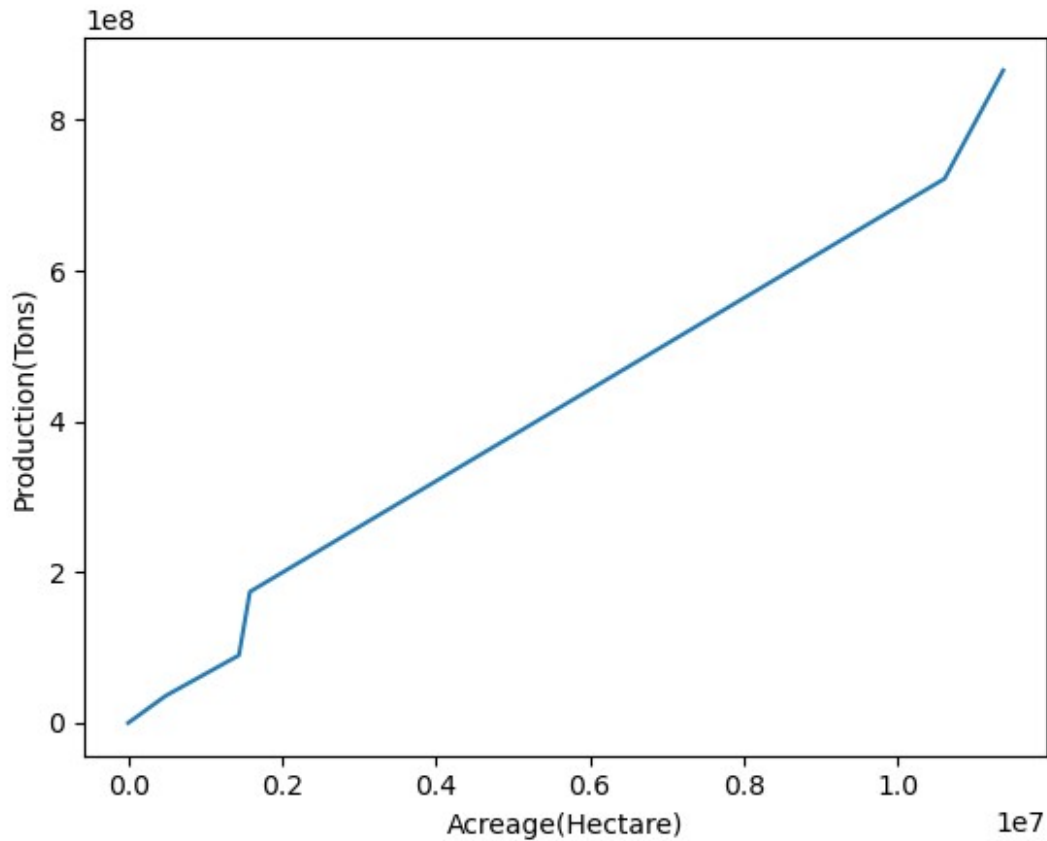
```
continent_names = df_continent.index.to_list()
sns.lineplot(data = df_continent, x = "number_of_countries", y=
"Production(Tons)" )
plt.xticks(df_continent["number_of_countries"], continent_names,
rotation =90)
plt.show()
```



Do continent with highest land produces more sugarcane?

```
sns.lineplot(data = df_continent, x = "Acreage(Hectare)", y =  
"Production(Tons)" )
```

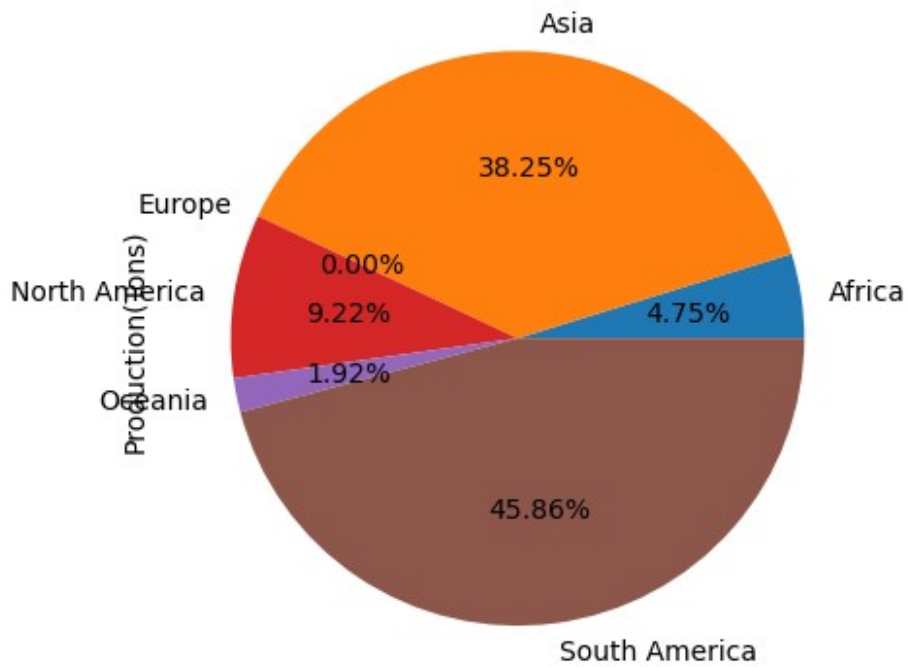
```
<Axes: xlabel='Acreage(Hectare)', ylabel='Production(Tons)'\>
```



Production distribution by continent

```
df_continent["Production(Tons)"].plot(kind = "pie", autopct = "%.2f%")
plt.title('Production Distribution by Continent')
plt.show()
```

Production Distribution by Continent



In addition understanding all the graph in one go
`sns.pairplot(df)`

`<seaborn.axisgrid.PairGrid at 0x220d9220620>`

